

E027

Virtualization and IaaS

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional bank is building a new virtualization cluster for production workloads. The team installs the hypervisor directly on the servers with no operating system underneath. The servers were delivered in April and have redundant power supplies. What type of hypervisor is this?

- A. Type 2 (hosted).
- B. Type 2, because redundant power supplies need a host OS to manage them.
- C. Type 1 (bare-metal).
- D. A container runtime.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A student installs virtualization software on her Windows laptop to run a Linux test VM. What type of hypervisor is she using?

- A. Neither; hypervisors cannot run on laptops.
- B. Type 2 (hosted).
- C. Type 1 (bare-metal).
- D. IaaS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A development team on Linux hosts needs to run a legacy application that only works on Windows Server, fully isolated from other tenants. The team lead prefers containers because they start in seconds, and the project has a \$20,000 budget. What should they use?

- A. A virtual machine running Windows Server.
- B. A container on the Linux host.
- C. Bare-metal Linux with no virtualization.
- D. A container, because fast startup matters most.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An agency is consolidating 60 physical servers that each use about 8 GB of RAM. New hosts have 256 GB of RAM, and policy keeps 25% of each host's memory free as headroom. The old servers are five years old and run a mix of Windows and Linux. How many hosts are needed, before any failover spare?

- A. 3
- B. 8
- C. 2
- D. It cannot be sized until the OS mix is known.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A cluster has four hosts with 512 GB of RAM each. One host is held back as a failover spare (N+1), and the others keep 20% headroom. Each VM needs 16 GB. How many VMs can the cluster run?

- A. 96
- B. 76
- C. 102
- D. 19

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A retailer runs its web servers on IaaS VMs from a provider with data centers on three continents. A security scan finds an outdated web server package on the guest OS. The provider publishes a monthly security newsletter. Who must update the package?

- A. Both equally.
- B. The retailer.
- C. The provider, since it runs the data centers.
- D. The provider, because its newsletter covers security.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A clinic takes nightly snapshots of its records VM on the same storage array that holds the VM's disks. The array is under a premium support contract and sits in a locked room. What is the main weakness of this protection plan?

- A. If the array fails, the VM and its snapshots are lost together; it needs independent backups stored elsewhere.
- B. The support contract should be upgraded.
- C. Snapshots should be taken hourly instead.
- D. None; snapshots are full backups.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

When is a VM snapshot the right tool?

- A. To protect against loss of the storage array.
- B. Always, instead of backups.
- C. Right before applying a patch, to roll back quickly if it fails.
- D. For seven-year legal retention.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A cluster has HA enabled but not fault tolerance. At 2 p.m. on a busy Tuesday, one host loses power unexpectedly. The cluster was set up by a contractor in 2023. What happens to the VMs on that host?

- A. They stay down until an administrator acts.
- B. They are live-migrated before the host fails.
- C. They keep running with no interruption.
- D. They go down and are restarted on other hosts after a brief outage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A host with 128 GB of RAM runs VMs allocated 200 GB in total, and during month-end most VMs actively use their full memory. The same host has 32 cores running VMs allocated 64 vCPUs. The host is in rack 14. Which resource is most likely causing month-end slowdowns?

- A. CPU, because 2:1 vCPU overcommit always causes slowdowns.
- B. Neither, since hypervisors share resources perfectly.
- C. Memory, because active use beyond physical RAM forces ballooning and swapping.
- D. Rack location.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A workload runs 24 hours a day all year. On-demand costs \$0.20 an hour; a one-year reserved instance costs \$1,051.20 in total. The team prefers Linux and deploys from a Git repository. How much does reserved save per year?

- A. -\$700.80 (reserved costs more)
- B. \$0.08
- C. \$1,752.00
- D. \$700.80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Using the same prices (\$0.20 an hour on-demand; \$1,051.20 for a one-year reservation), how many hours a year must the workload run for reserved to break even?

- A. Any amount; reserved is always cheaper.
- B. 8,760 hours
- C. 3,504 hours
- D. 5,256 hours (about 60% of the year)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A regional bank is building a new virtualization cluster for production workloads. The team installs the hypervisor directly on the servers with no operating system underneath. ~~The servers were delivered in April and have redundant power supplies.~~ What type of hypervisor is this?

Target word: classify the hypervisor **Decides:** No host OS underneath.

A	Inverted operation	Type 2 runs on a host OS.
B	Camouflage	Power hardware does not decide the hypervisor type.
C	Correct	Runs directly on hardware.
D	Confusion set	Containers share an OS kernel; they are not hypervisors.

2. Answer B Concept 1 • transfer

A student installs virtualization software on her Windows laptop to run a Linux test VM. What type of hypervisor is she using?

Target word: reverse: host OS present **Decides:** Runs on top of Windows.

A	Absolute language	Hosted hypervisors commonly do.
B	Correct	Hosted on a desktop OS.
C	Inverted operation	Would need no host OS.
D	Adjacent category	A cloud service model, not a hypervisor type.

3. Answer A Concept 2

A development team on Linux hosts needs to run a legacy application that only works on Windows Server, fully isolated from other tenants. The team lead prefers containers because they start in seconds, ~~and the project has a \$20,000 budget.~~ What should they use?

Target word: VM vs. container **Decides:** Needs a different OS kernel and strong isolation.

A	Correct	VMs run their own OS with strong isolation.
B	Inverted operation	Containers share the host kernel; a Linux host cannot run a Windows-kernel container.
C	Dropped element	Drops the Windows requirement.
D	Camouflage	Preference for speed does not overcome the OS requirement.

4. Answer A Concept 3

An agency is consolidating 60 physical servers that each use about 8 GB of RAM. New hosts have 256 GB of RAM, and policy keeps 25% of each host's memory free as headroom. ~~The old servers are five years old and run a mix of Windows and Linux.~~ How many hosts are needed, before any failover spare?

Target word: host count **Decides:** Usable $192 \text{ GB} \div 8 \text{ GB} = 24$ VMs per host; $60 \div 24 = 2.5 \rightarrow 3$.

A	Correct	$256 \times 0.75 = 192 \text{ GB}$; 24 VMs per host; round 2.5 up to 3.
B	Inverted operation	Treats the 25% headroom as the usable share.
C	Dropped element	Ignores headroom: $256 \div 8 = 32$ VMs per host.
D	Camouflage	Memory demand is given; OS mix does not decide.

5. Answer B Concept 3 · transfer

A cluster has four hosts with 512 GB of RAM each. One host is held back as a failover spare (N+1), and the others keep 20% headroom. Each VM needs 16 GB. How many VMs can the cluster run?

Target word: solve for VM count **Decides:** $3 \times 512 \times 0.8 = 1,228.8 \text{ GB} \div 16$.

A	Dropped element	Leaves out the 20% headroom.
B	Correct	$1,228.8 \div 16 = 76.8$; round down to 76 whole VMs.
C	Adjacent category	Counts the failover spare as usable capacity.
D	Inverted operation	Uses the 20% headroom as the usable share.

6. Answer B Concept 4

A retailer runs its web servers on IaaS VMs from a provider ~~with data centers on three continents~~. A security scan finds an outdated web server package on the guest OS. ~~The provider publishes a monthly security newsletter~~. Who must update the package?

Target word: IaaS responsibility **Decides:** Software on the guest OS is the customer's.

A	Confusion set	Shared responsibility splits layers; it does not share each task.
B	Correct	Customers manage the OS and applications in IaaS.
C	Adjacent category	Data centers and hypervisor, not guest software.
D	Camouflage	A newsletter does not transfer responsibility.

7. Answer A Concept 5

A clinic takes nightly snapshots of its records VM on the same storage array that holds the VM's disks. ~~The array is under a premium support contract and sits in a locked room~~. What is the main weakness of this protection plan?

Target word: snapshot limitation **Decides:** Snapshots depend on the same storage.

A	Correct	Snapshots are not separate copies.
B	Camouflage	Support does not create an independent copy.
C	Dropped element	More snapshots on the same array share the same failure.
D	Confusion set	They depend on the original disks.

8. Answer C Concept 5 · transfer

When is a VM snapshot the right tool?

Target word: reverse: proper snapshot use **Decides:** Short-term rollback point.

A	Adjacent category	Snapshots live on that array.
B	Absolute language	They complement, not replace, backups.
C	Correct	Short-term, point-in-time rollback.
D	Inverted operation	Long-term retention needs backups.

9. Answer D Concept 6

A cluster has HA enabled but not fault tolerance. At 2 p.m. ~~on a busy Tuesday~~, one host loses power unexpectedly. ~~The cluster was set up by a contractor in 2023.~~ What happens to the VMs on that host?

Target word: HA behavior **Decides:** HA restarts after failure.

A	Inverted operation	HA restarts them automatically.
B	Adjacent category	Live migration needs a running source host; failure is unplanned.
C	Confusion set	That is fault tolerance.
D	Correct	HA restarts; it does not keep VMs running through failure.

10. Answer C Concept 7

A host with 128 GB of RAM runs VMs allocated 200 GB in total, and during month-end most VMs actively use their full memory. The same host has 32 cores running VMs allocated 64 vCPUs. ~~The host is in rack 14.~~ Which resource is most likely causing month-end slowdowns?

Target word: risky overcommit **Decides:** Active memory use exceeds physical RAM.

A	Absolute language	Moderate CPU overcommit is common and usually fine.
B	Inverted operation	Sharing cannot create memory that does not exist.
C	Correct	Memory overcommit hurts when demand is real.
D	Camouflage	Position does not decide performance.

11. Answer D Concept 8

A workload runs 24 hours a day all year. On-demand costs \$0.20 an hour; a one-year reserved instance costs \$1,051.20 in total. ~~The team prefers Linux and deploys from a Git repository.~~ How much does reserved save per year?

Target word: annual savings **Decides:** On-demand = $\$0.20 \times 8,760 = \$1,752$.

A	Inverted operation	Subtracts in the wrong direction.
B	Adjacent category	The hourly gap ($\$0.20 - \0.12), not the annual saving.
C	Dropped element	Leaves out the reserved cost.
D	Correct	$\$1,752 - \$1,051.20 = \$700.80$.

12. Answer D Concept 8 • transfer

Using the same prices (\$0.20 an hour on-demand; \$1,051.20 for a one-year reservation), how many hours a year must the workload run for reserved to break even?

Target word: solve for break-even hours **Decides:** $\$1,051.20 \div \0.20 .

A	Absolute language	Below break-even, on-demand wins.
B	Wrong stem number	Hours in a full year.
C	Inverted operation	$8,760 - 5,256$; the hours not run.
D	Correct	$\$1,051.20 \div \$0.20 = 5,256$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Type 1 vs. Type 2 hypervisors	Type 1 (bare-metal) runs directly on hardware and is used in data centers and IaaS; Type 2 (hosted) runs on top of a desktop OS for labs and testing.	Calling a hypervisor installed on Windows or macOS “bare-metal.”	Whether there is a host operating system under the hypervisor.
2	VMs vs. containers	VMs virtualize hardware and each runs its own OS; containers share the host kernel, start faster, and use less overhead, with weaker isolation.	Assuming containers can run a different OS kernel than the host.	Need for a different OS, strong isolation, or fast startup density.
3	Consolidation and capacity sizing	Hosts needed = total demand ÷ usable capacity per host, where usable capacity leaves headroom (and a failover host if required); always round up.	Sizing to 100% of host capacity or forgetting the failover host.	Per-VM demand, host size, and a headroom or N+1 requirement.
4	IaaS responsibility	In IaaS the provider runs hardware, storage, network, and hypervisor; the customer runs the guest OS, middleware, apps, and data.	Expecting the provider to patch the guest OS.	A task in the guest OS or above.
5	Snapshots are not backups	A snapshot is a point-in-time state that depends on the original disk and storage; a backup is an independent copy stored separately.	Relying on snapshots on the same storage as a backup.	Storage failure or long-term retention.
6	Live migration vs. HA vs. fault tolerance	Live migration moves a running VM with no downtime for planned work; HA restarts VMs on another host after a failure (brief outage); fault tolerance runs a shadow copy for zero downtime.	Thinking HA gives zero downtime.	Planned maintenance vs. unexpected host failure.
7	Overcommitment	CPU overcommit is common and safe within limits; memory overcommit is riskier because active use beyond physical RAM causes ballooning and swapping that slow every VM.	Treating memory like CPU for overcommit.	Allocated resources exceed physical ones.
8	IaaS pricing models	On-demand is flexible and highest per hour; reserved is cheaper for steady long-running use; spot is cheapest but can be reclaimed. Reserved breaks even at reserved cost ÷ on-demand hourly rate.	Putting interruptible or rarely-run workloads on reserved capacity, or critical steady ones on spot.	Usage pattern plus prices per hour and per term.

Course scope (public description): Virtualization and Infrastructure as a Service (IaaS) explores the foundational principles and benefits of virtualization technologies and their application in IaaS environments. This course examines the role of virtualization in modern IT infrastructure, emphasizing its practical implementation and contribution to scalable, efficient, and cost-effective solutions.